A panel obstruction in an omitted Mathieu geometry

An addition to the residue shadow calculations

The rank-three M_{12} entry in Tables 1 and 7 of Jacobs and Leemans [1] gives eighteen eligible geometries, divided into seven satisfying (IP), one satisfying (WIP) alone, and ten failing (WIP). Eligibility includes firmness, flag transitivity, residual connectedness, residual weak primitivity, local 2-transitivity, and geometric $(\mathrm{IP})_2$. Our earlier calculation on the twenty active rank-three inputs in the supplied archive found seventeen eligible geometries, with counts $7, 1, 9$. Three fail local 2-transitivity.

There is a separate potential omission. The retained complete subgroup census contains twenty-one rank-three classes under automorphisms of the specified acting group and permutations of types. The exact comparison with every raw archive family identifies twenty of them; class 18 of that census is absent. This index belongs to our census, not to the historical numbering in the paper. We determine whether this additional class supplies an eighteenth eligible geometry.

Proposition 1. *Let $G = M_{12}$ and let (H_1, H_2, H_3) be the marked subgroup family in retained census class 18. Its orders are*

$$|G| = 95040, \qquad (|H_1|, |H_2|, |H_3|) = (24, 660, 660), \qquad |H_1 \cap H_2 \cap H_3| = 6.$$

The geometry satisfies all of the above eligibility conditions except local 2-transitivity. It satisfies the full geometric intersection property. Its panel of type 1 is the action of A_5 on the ten two-element subsets of a five-element set. In particular, it cannot be added to the locally 2-transitive population.

Proof. The entire marked family is retained in `prior_representatives.g`. The unchanged exact residue procedure from the common shadow paper checks every base residue, including panels, and every remaining type. Flag transitivity makes these representatives exhaustive for residues of the same type. The resulting certificate verifies firmness, residual connectedness, residual weak primitivity, and the geometric intersection property in ranks two and three. These are exact finite group and coset calculations; no property is inferred solely from the displayed orders.

The failure of local 2-transitivity has a smaller independent proof. Put $B = H_1 \cap H_2 \cap H_3$ and $K = H_2 \cap H_3$. A bijective homomorphism $K \longrightarrow A_5$ maps B to

$$D = \langle (2\,4\,3), (1\,5)(2\,3) \rangle.$$

The program `identify_panel.g` checks this homomorphism and that D is exactly the stabilizer of $\{1, 5\}$. Thus the right B-cosets in K are identified equivariantly with the ten two-element subsets. This also gives a proof that is independent of the shadow procedure: `panels.g` constructs those ten right cosets directly and computes the action of their point stabilizer.

The stabilizer D has three orbits on the two-element subsets. The fixed subset is $\{1, 5\}$. There are three subsets disjoint from it, and six meeting it in one point. The displayed three-cycle permutes the complementary three points, while the displayed involution interchanges 1 and 5 and two of those three points. These generators act transitively on each of the latter two sets. The subdegrees are therefore $1, 3, 6$. Since the point stabilizer is not transitive on the other nine points, the panel action is not 2-transitive. The other two panels each have degree two. This proves the stated failure without relying on a previously recorded eligibility flag. □

Consequently, relative to the retained complete census and its exact archive correspondence, there are seventeen eligible rank-three M_{12} classes and the shadow counts remain $7, 1, 9$. This closes the possible missing-input explanation of the earlier discrepancy. It does not identify any class with a historical geometry number in Tables 2 or 8, nor does it justify changing the stated equivalence convention. The correspondence of historical labels remains a separate obligation in a complete recovery of the paper.

The census, its subgroup catalogue and its completeness proof are retained premises. The common residue theorem and its whole program are also retained; they must be acquired in any comparison using this application. The new calculation does not repeat the census. No new original-paper recovery, open-conjecture result or Lean formalization is claimed here. Software remains outside manuscript length comparisons on both sides.

References

[1] P. Jacobs and D. Leemans, *An algorithmic analysis of the intersection property*, LMS Journal of Computation and Mathematics **7** (2004), 284–299.

[2] *Shadows and residues in finite coset geometries*, the retained six-page common proof and its complete residue program.